

Technical Report per Attachment B1 of the General Order

General Waste Discharge Requirements for Small Domestic
Wastewater Treatment System General Information for Discharger

November 2022

Prepared by:
California Rural Water Association



California
Rural Water Association

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D. R. Lafontaine

Applicant:
Madera County Maintenance District No. 28 Ripperdan Wastewater
Treatment Facility (WWTF)

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1.0 BACKGROUND

1.1 Wastewater System description

1.1.1 Description of wastewater system and how wastewater is generated

Madera County Maintenance Division No. 28 – Ripperdan WWTF is in the community of Ripperdan CA located South of Madera on Hwy 145. The Ripperdan subdivision was built in 1970's. The collection system routes wastewater from 16 homes (8 on each side of the treatment system) to the WWTF in the center of the development. The WWTF consists of a wet well, extended aeration bed, clarifier. Waste is either routed to one of 16 leach pits or if needed a holding tank. The Ripperdan WWTF is currently operating under Waste Discharge Requirement order number 97-10-DWQ which was updated in 2014.

1.1.2 Site Location Map and Site Plan

See Vicinity Map and Site Map as attachments to this technical report (Figure 1 and 2, respectively).

1.1.3 WWTF – Groundwater wells and surface water bodies information and location

One groundwater well exists in the WWTF compound in the Northeast corner. The well has been included in the site plan.

The closest surface water body is the San Joaquin River located approximately two (2) miles South of Ripperdan.

1.1.4 Assessor's Parcel Number, section number and Township and Range

The WWTF is located South of Madera along highway 145 in the center of the subdivision of Ripperdan. All components are housed in the wastewater treatment compound. All treatment components are outside or buried in the ground, there are no wastewater system buildings onsite.

The WWTF and leach pits are located at parcel number 040-230-009, Section 25, Town 12S, Range 17E.

1.1.5 Describe the Water Supply to the residence(s), business(es), and or other facilities being served by the wastewater system

Water is supplied to 16 homes surrounding the Ripperdan WWTF by one groundwater well located inside the WWTF compound. The groundwater well was reported by operators onsite to be a total depth of 300 feet and screened from 198 to 220 feet below ground surface (bgs). Water is stored in a water tank onsite prior to being gravity fed to the houses.

1.2 Service Area Description

1.2.1 Describe the proximity of the wastewater system to an existing regional collection system

The City of Madera WWTF is located on Road 21 ½ and Avenue 13 approximately 5.7 miles North North-West of the Ripperdan WWTF.

1.2.2 Wastewater collection system. For collection system subject to State Water Board Order No. 2006-0003-DWQ, reports prepared to comply with the Order satisfies this requirement and may be submitted as part of the ROWD technical report.

The collection system is approximately 750 feet (0.15 miles).

1.2.2.1 Age and condition of collection system

The collection system was built in the 1970's and is in good condition.

1.2.2.2 Piping construction and layout (show on map)

The collection system map has been included as Figure 3.

1.2.2.3 Lift Station and backup pumping systems

The collection system has one wet well in the WWTF compound.

1.2.2.4 Failure warning system

The wet well is equipped with a high-level alarm with a flashing light and audible horn. There are no alarms in the collection system.

1.2.2.5 Inflow and infiltration (I&I) Estimates

The chief plant operator indicates that I & I is not a problem in the Ripperdan collection system. The operators estimate that I & I account for less than 10% to the normal dry flow.

1.2.2.6 Maintenance of Collection system and spill response

If the wet well backs up there is one manhole to the South that backs up after 24 hours. Operators have spill kits and sandbags on each truck should a Sanitary Sewer Overflow (SSO) occur. The operators indicate that they hydro-clean the collection lines every other year.

1.2.3 Storm Water collection System

1.2.3.1 Storm Water Collection area (Show on map)

There is a stormwater collection system throughout Ripperdan.

1.2.3.2 Storm water disposal area in relation to wastewater disposal area.

Unknown

1.2.3.3 Storm water disposal permit

Unknown.

2.0 WASTEWATER CHARACTERIZATION AND TREATMENT

2.1 Domestic wastewater characterization (untreated wastewater)

2.1.1 Describe the generation of wastewater (retirees, families, recreational vehicles [RV] institution, etc.)

Wastewater is generated from the development of Ripperdan a 16-home community that services majority of year-round residential units.

2.1.2 Domestic wastewater flow rate. Describe any special events or seasonal variations that cause high wastewater flow rates or other sources of wastewater

The treatment plant was designed for an influent flowrate of 7,500 gallons per day (0.0075 MGD) and a population of about 64 people. The system has no flow meter and flow rates are estimated off pump run times. Present flowrates average 1,126 gallons per day in 2021 (Appendix 1). There are no special events or seasonal variations.

2.1.3 Characterize domestic wastewater for Biological Oxygen Demand (BOD), Total suspended Solids (TSS), electrical conductivity, nitrogen, sodium, chloride, and specific constituents of concern as needed based on site activities. Characterize wastewater for holding tank chemicals identified in the General Order

Influent samples are collected for BOD and TSS. Samples are collected quarterly and sent to Moore Twining Associates, Inc a California ELAP Certificate #1371 Laboratory for analysis. Sample results for 2021 are listed below and included as Appendix 2:

Table 1: Influent Sample Collection Results for 2021

Sample Date(s)	BOD (mg/L)	TSS (mg/L)
01/21/21	110	130
04/07/21 and 04/13/21	160	530
09/29/21 and 10/04/21	190	330
11/10/21	230	250

2.2 Wastewater treatment system

2.2.1 Provide a wastewater treatment schematic

A process and flow diagram (PFD) has been included as Figure 4.

2.2.2 Describe wastewater pretreatment components

All wastewater is routed from residences through the collection system comes to a 20-foot deep wet well with two pumps capable of pumping at 100 gallons per minute each.

2.2.2.1 Domestic wastewater pretreatment system (e.g., septic tank, effluent pump system, grease traps, etc.)

N/A

2.2.2.2 Describe storage, treatment, and disposal of pretreatment residuals

N/A

2.2.3 Describe preliminary treatment activities (e.g., screening, comminution, grit removal)

The WWTF has no screening.

2.2.3.1 Describe storage, treatment, and disposal of preliminary treatment residuals

N/A

2.2.4 Describe primary treatment activities (remove settleable/floatable matter)

Water is pumped from the wet well to a settling/trash tank where grit falls to the bottom. Operators remove any debris or grit in the holding well as needed.

2.2.4.1 Describe storage, treatment, and disposal, and disposal of treatment residuals

Screenings and debris are removed from the settling/trash tank and placed into a 55-gallon drum onsite. Once full the 55-gallon drum is taken and disposed of at an appropriate landfill.

2.2.5 Describe the treatment technology

Primary treatment activities are performed in an extended aeration basin. The aeration basin is 10 feet deep and has a surface area of 10 feet across by 24 feet long. The extended aeration basin has a volume of 2,400 cubic feet (approximately 18,000 gallons). Scum is collected on top in a sludge collection box. After the aeration basin the wastewater is routed to a final tank (where solids fall to the bottom) prior to disposal.

2.2.5.1 Describe storage, treatment, and disposal of treatment residuals (e.g., sludge, septage, etc.)

The extended aeration unit produces sludge in quantities that require little removal. Operators monitor sludge depth and will remove the sludge, should it be required. Operators remove scum from the

sludge collection box and combine it with the primary grit in a 55-gallon drum prior to disposal.

2.2.6 Size and location of treatment equipment (e.g., septic tank volume, package treatment plant, membrane biological reactor, pond size include acreage and storage capacity, pond liners, and number and horsepower of aerators, etc.)

The Package plant is sized for an influent of 7,500 gallons per day. Dimensions of the aeration basin and clarifier were determined during an onsite visit. Storage capacities are determined from the measurements as shown below:

Table 2: Treatment System Capacity

Equipment	Length (ft)	Width (ft)	Height (ft)	Volume (ft ³)	Volume (gal)
Settling tank	8	10	10	800	5,984
Extended Aeration Basin	24	10	10	4,000	17,952
Clarifier	8	10	10	800	5,984
Holding tank					5,000
Total Storage of System					34,920

The extended aerator has a rotary lobe pump assumed to be 2-horsepower (nameplate not legible) which supplies air to 3 diffusers plumbed into the extended aeration basin. Two additional diffusers are plumbed into the holding tank and the clarifier. The additional diffusers are not presently used.

2.2.7 Disinfection system equipment

N/A

2.2.8 Storage facilities

2.2.8.1 If wastewater will be stored prior to disposal the size and location of wastewater storage ponds including a map showing all the ponds and describe then as lined or not. Describe the materials, age, and condition of any liners.

Wastewater is routed to holding tank after final clarification. The holding tank is a poly tank that has as estimated capacity of 5,000 gallons (based on dimensions of the tank estimated at 10-foot diameter by 8-foot height).

2.2.9 Predicted wastewater effluent quality

2.2.9.1 Characterize the wastewater for TSS, BOD, total coliform organisms (if needed) and specific constituents of concern as needed. If RV waste is discharged to the system, characterize for holding tank chemicals identified in the General Order

Effluent samples are collected for BOD and TSS. Samples are collected quarterly and sent to Moore Twining Associates, Inc a California ELAP Certificate #1371 Laboratory for analysis. Sample results for 2021 are listed below and included as Appendix 2:

Table 3: Effluent Sample Results for 2021

Sample Date	BOD (mg/L)	EC (μS/cm)	Nitrate as N (mg/L)	Nitrite as N (mg/L)	pH (pH units)	TKN (mg/L)	Total Nitrogen (mg/L)	TSS (mg/L)
01/21/21	3.5	660	<0.90	<0.60	7.8	15	15	<4.0
04/07/21	3.2	630	<0.90	<0.60	7.9	15	15	<4.0
09/28/21	14	710	<0.80	<0.60	7.9	16	16	12
11/09/21	3.9	700	1.8	<0.60	7.8	6.8	8.5	<4.0

RV waste is not allowed into the system and therefore holding tank chemicals are not sampled for.

2.2.10 Treated effluent Disposal

2.2.10.1 Describe how treated wastewater will be dispersed (land application area, leach field, percolation pond)

Wastewater effluent is either routed to the holding tank or gravity fed into one of 16 leach pits. The leach pits have dimensions of 4-feet in diameter and have a depth of 35-feet. The 16 leach pits volume is calculated in Table 3 below:

Table 4: Leach Pit Volume Calculation

Leach Pit	Diameter (ft)	Cross Sectional Area (ft ²)	Depth (ft)	Volume (ft ³)	Volume (gallons)
Each	5	19.6	35	686	5,131
Maximum storage Capacity of 16 Leach Pits					82,100

2.2.10.2 Describe the proposed area (and the 100-percent replacement area when needed, such as for a leach field disposal system) include acreage, surrounding land use, depth to groundwater, and the proximity of drainage ways, surface waters, and municipal, industrial, or agricultural wells

The depth to groundwater was reported by operators to be approximately 198 to 220 feet below ground surface (bgs).

Surrounding land uses are primarily agricultural with a small light commercial center South of the site.

The closest surface water body is the San Joaquin River located approximately two (2) miles South of Ripperdan.

One municipal groundwater well exists in the WWTF compound in the Northeast corner. The well has been included in the site plan.

2.2.10.2.1 If land application is proposed, describe how storm water that falls on the land application area (LAA) is handled. If storm water is allowed to run off the LAA, contact your Regional Water Board representative to discuss wastewater disinfection requirements

Land application is not being used wastewater is routed to 16 leach pits.

2.2.10.2.2 Provide a water balance that demonstrates adequate storage/disposal capacity. Identify the safety factors used in the calculations

A water balance is provided as Appendix 3 to this technical report.

2.2.10.2.3 Support the assumptions and calculations in the water balance with adequate information

Average monthly flowrates are collected from the monthly records. Leach pit volumes are calculated from the volume when they were constructed. Percolation in seepage/leach pits occur primarily through the side walls and not the bottom of the pit which is clogged by sediment from the treated wastewater. Percolation will be assumed to be occurring predominantly from the side walls. The surface area of each seepage/leach pit (Pit SA) assuming no percolation occurs through the bottom (clogged by wastewater sediment) and the top (no percolation occurs upward) is calculated as the following:

$$\begin{aligned} Pit\ SA &= 2 * \pi * r * h \\ \text{Where } r &= 2\text{ft and } h = 35\text{ ft} \\ Pit\ SA &= 2 * \pi * 2\text{ (ft)} * 35\text{ (ft)} \\ Pit\ SA &= 440\text{ (ft}^2\text{)} \end{aligned}$$

The surface area of all 16 seepage/leach pits is 7,040 ft². Based on WDR No 90-261, Soils onsite consist of Tujunga sandy loam and Hanford fine loamy sand. Being conservative Very fine sandy loam is reported to have an infiltration rate of

1.02 ft/day (as reported by *A Field Method for Measurement of Infiltration* for the Department of the Interior 1963, Table 1) below:

FIELD METHOD FOR MEASUREMENT OF INFILTRATION F-7

TABLE 1.—Infiltration rates for different type soils as measured by infiltrometer rings in third hour of a wet run

[After Free, Browning, and Musgrave, (1940)]

Soil type	Porosity (percent)		Infiltration rate		
	Total	Non-capillary (specific yield)	Inches per hour	Feet per day	Centimeters per hour
Gravelly silt loam.....	54.9	28.1	4.96	9.92	12.60
Clay loam.....	61.1	36.3	3.98	7.96	10.11
Silt loam.....	57.0	32.0	2.09	4.18	5.31
Sandy loam.....	49.6	26.3	1.93	3.86	4.90
Clay (eroded).....	54.3	28.7	1.78	3.56	4.52
Sandy clay loam.....	48.8	27.7	1.42	2.84	3.61
Silty clay loam.....	50.8	24.3	.72	1.44	1.83
Stony silt loam.....	59.7	32.6	.55	1.10	1.40
Fine sandy loam.....	41.5	24.2	.55	1.10	1.40
Very fine sandy loam.....	49.6	23.4	.51	1.02	1.29
Loam.....	45.7	17.2	.50	1.00	1.27
Sandy clay.....	42.9	16.9	.05	.10	.13
Heavy clay.....	57.8	27.0	.02	.04	.05
Light clay.....	47.0	19.8	.00	.00	.00
Clayey silt loam.....	49.4	17.6	.00	.00	.00

The assumption is also made that the seepage/leach pits are not full and operate with at least 25% capacity at all times (conservative).

2.2.10.2.4 The use of subsurface disposal including leach fields and/or seepage pits serving more than 20 people, or systems that accept non-sanitary waste (generated by manufacturing, contains biocidal agents such as RV or portable toilets, etc.) must comply with the United States Environmental Protection Agency Underground Injection Control requirements

Based on the number of houses the WWTF accepts waste from 16 homes. The homes are mostly families and have an

assumed average occupancy of 4 people per household. Based on the number of homes and the average occupancy the WWTF serves approximately 64 people thus the system must comply with the United States EPA Underground Injection Control requirements. Based on the size and type of waste the only requirement at this time is to notify the Region 9 office by submitting a UIC Injection Well Inventory Form online. The UIC Injection Well Inventory Form was submitted when the WWTF was installed. The wells are registered under EPA number CA530085.

2.3 Recycled Water Projects

2.3.1 If treated wastewater will be applied for applied for beneficial uses, provide a title 22 Engineering report and the State Water Board Division of Drinking Water review/approval letter

Treated wastewater will not be applied for beneficial uses.

2.4 Operation and Maintenance

2.4.1 Describe routine operation and maintenance procedures

Operation and maintenance are performed 3 days a week when the operator(s) is(are) onsite. Operators record flows (pump hours), check dissolved oxygen, check for signs of ponding in the pit area. Weed abatement is performed when needed around the compound.

2.4.2 Treatment Operator Training and Qualifications Requirements

The operator on record is Jason Mitchell who has a wastewater operator certification grade II (certificate number 28676).

2.4.3 Contingency Plans for Repairs/Spills/Treatment issues

Repairs/spills are mitigated by the operators. All operators have spill kits and sandbags in their truck to mitigate spills back into the collection system. The county has access to a hydro flusher to clean lines as needed. A trailer mounted vacuum truck is also available in an emergency if needed.

3.0 GROUNDWATER QUALITY

3.1. Depending upon the threat to groundwater quality, groundwater monitoring wells maybe required

Groundwater monitoring wells are not present currently.

4.0 CALIFORNIA ENVIROMENTAL QUALITY ACT (CEQA)

4.1 Some existing small Domestic Systems will be determined to be categorically exempt from the CEQA act under Title 14, section 15301 (ongoing existing projects)

WWTF has been in operation since 1970's. Project is categorically exempt.

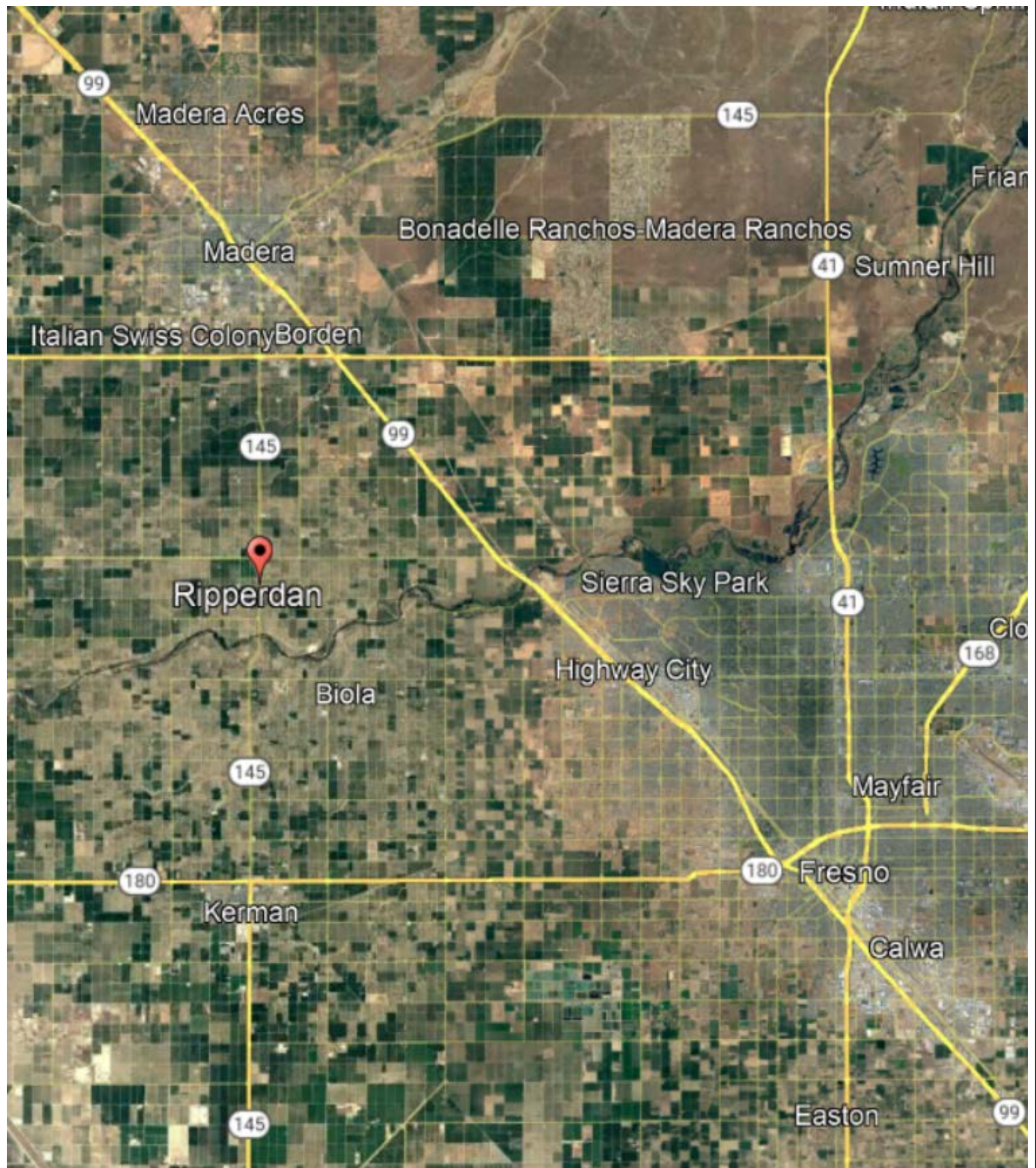
4.2. New or expanding Small Domestic System will likely require CEQA evaluation that should be performed by local agencies

N/A

5.0 ADDITIONAL TECHNICAL REPORTS

5.1.1 If required by the General Order, a Sludge Management Plan shall be submitted with the ROWD.

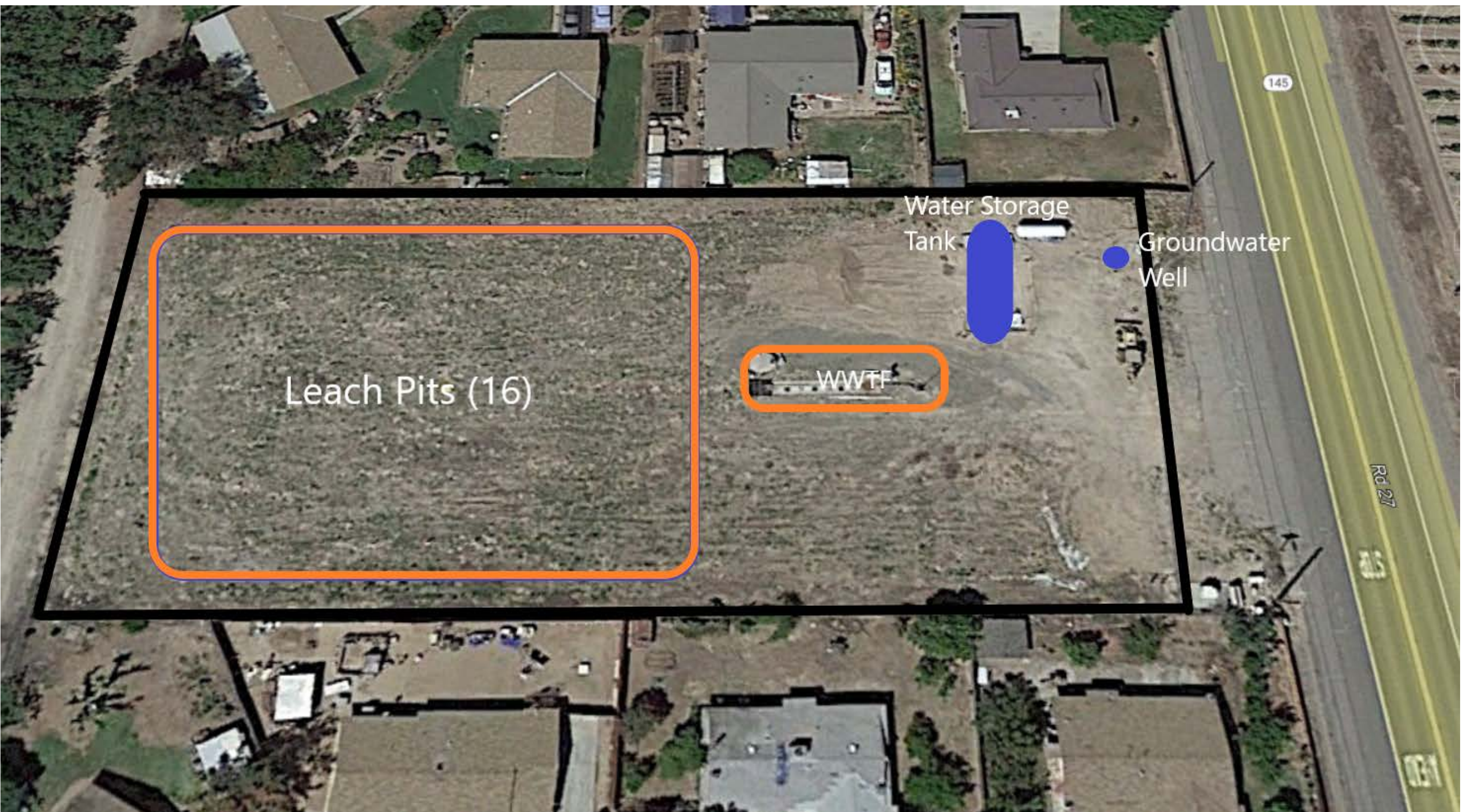
N/A



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VICINITY MAP

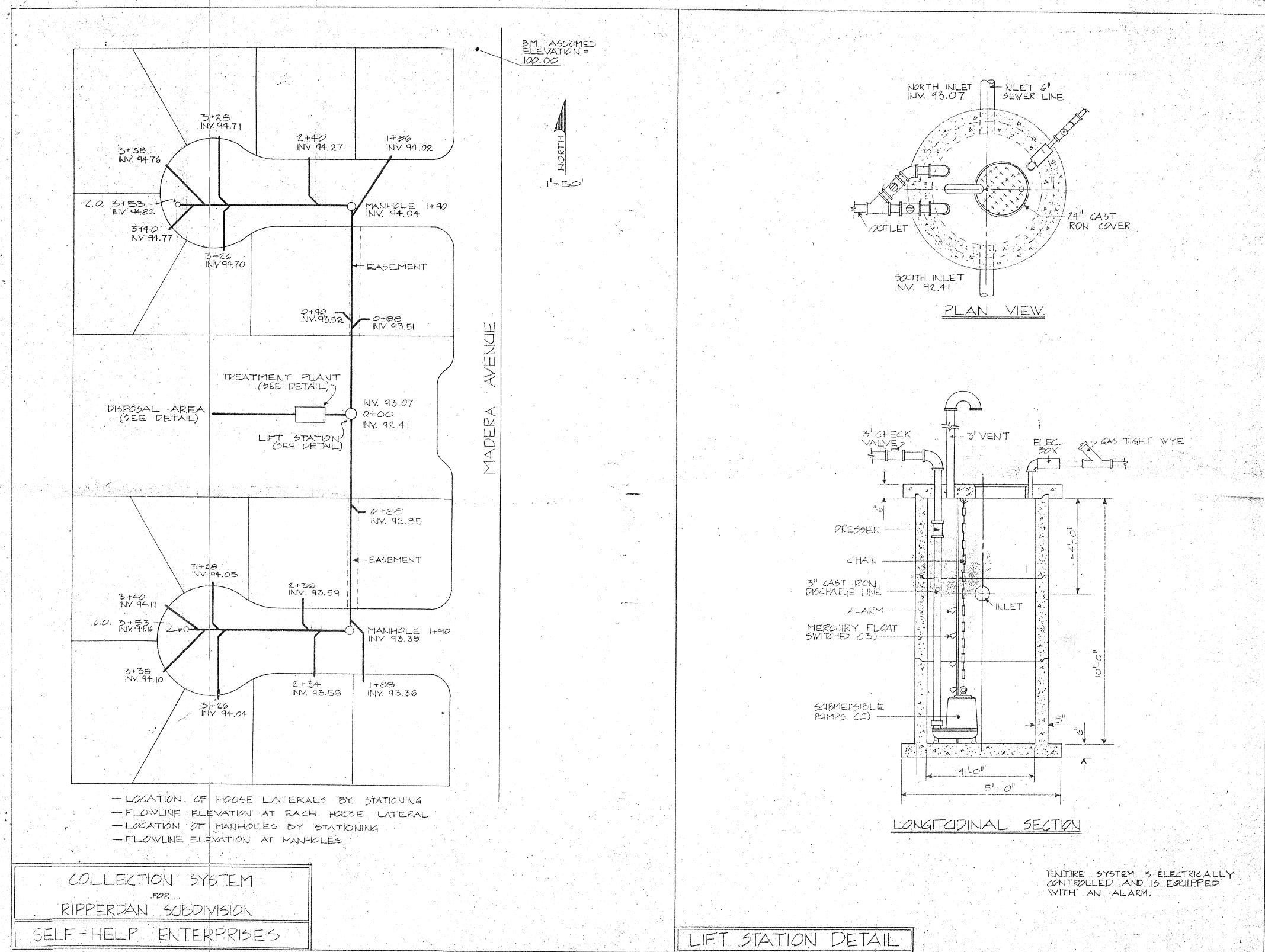
FIGURE
1

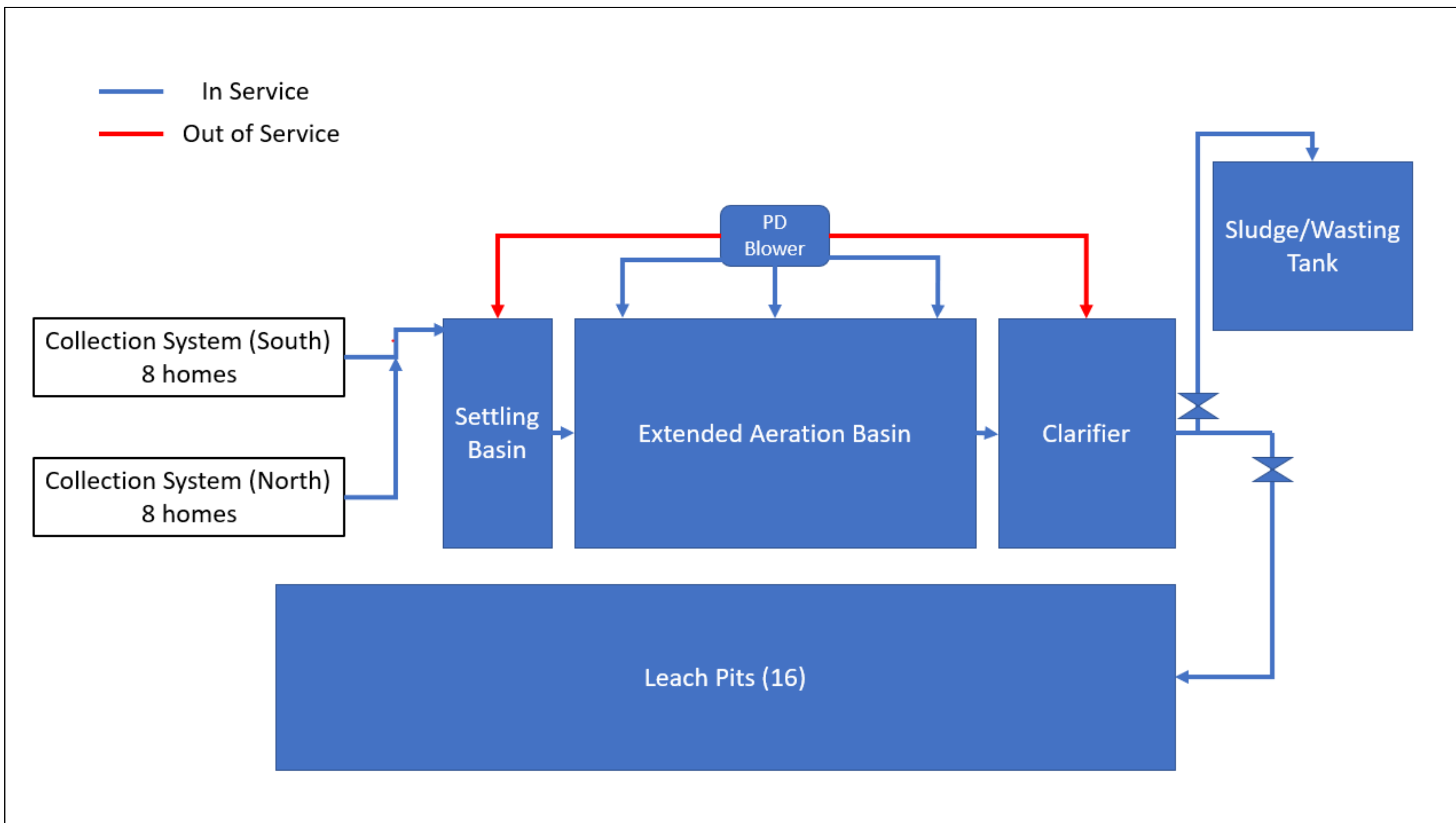


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SITE MAP

FIGURE
2





APPENDIX 1 – WWTF FLOWS 2021

Source	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Total Production
Plant Influent	41,900	34,700	43,500	30,900	42,700	35,400	27,200	21,300	14,400	26,800	23,300	62,300	404,400
													0
													0
													0
													0
													0
													0
Imported or Trucked In	0	0	0	0	0	0	0	0	0	0	0	0	0
District Totals	41,900	34,700	43,500	30,900	42,700	35,400	27,200	21,300	14,400	26,800	23,300	62,300	404,400

MD 285

APPENDIX 2 – LABORATORY ANALYTICAL DATA 2021

Madera County Public Works - Special Districts
 200 W. 4th Street, 3rd Floor
 Madera CA, 93637

Project: MD-28S Ripperdan
Project Number: System Number 2014-0820
Project Manager: Marty Duvall

Reported:
 02/04/2021

Plant EFF. (MD-28S)

HA21008-01 (Waste Water)

Sampled: 01/21/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		3.5	3.0	mg/L	3	B1A2112	01/21/21	01/26/21	SM 5210 B
Specific Conductance (EC)		660	1.0	µS/cm	1	B1A2122	01/21/21	01/21/21	SM 2510 B
Nitrate as N	D2	ND	0.90	mg/L	2	B1A2205	01/22/21	01/22/21	EPA 300.0
Nitrite as N	D2	ND	0.60	mg/L	2	B1A2205	01/22/21	01/22/21	EPA 300.0
pH		7.8	0.10	pH Units	1	B1A2122	01/21/21	01/21/21	SM 4500-H B
Total Kjeldahl Nitrogen		15	0.50	mg/L	1	B1A2202	01/22/21	01/25/21	EPA 351.2
Total Nitrogen		15	0.90	mg/L	2	[CALC]	01/25/21	01/25/21	(CALC)
Total Suspended Solids		ND	4.0	mg/L	1	B1A2506	01/25/21	01/25/21	SM 2540 D

Plant INF. (MD-28S)

HA21008-02 (Waste Water)

Sampled: 01/21/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		110	30	mg/L	30	B1A2121	01/21/21	01/26/21	SM 5210 B
Total Suspended Solids		130	20	mg/L	5	B1A2506	01/25/21	01/25/21	SM 2540 D

SS # Main Well

HA21008-03 (Ground Water)

Sampled: 01/21/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Specific Conductance (EC)		500	1.0	µS/cm	1	B1A2610	01/26/21	01/26/21	SM 2510 B

Notes and Definitions

- BS2 Recovery for this analyte was biased low. Results were accepted based on duplicate results.
- D2 Sample diluted due to high concentration(s) of non-target analyte(s).
- RPD3 The RPD is out of range for this spike and its duplicate due to a low or high bias of one of the two spikes.
- µg/L micrograms per liter (parts per billion concentration units)
- mg/L milligrams per liter (parts per million concentration units)
- mg/kg milligrams per kilogram (parts per million concentration units)
- ND Analyte NOT DETECTED at or above the reporting limit
- RPD Relative Percent Difference
- Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field.
 If the test was performed in the laboratory, the hold time was exceeded. (for aqueous matrices only)

Madera County Public Works - Special Districts
 200 W. 4th Street, 3rd Floor
 Madera CA, 93637

 Project: MD-28S Ripperdan
 Project Number: System Number 2014-0820
 Project Manager: Marty Duvall

 Reported:
 04/21/2021

Plant EFF. (MD-28S)

HD07011-01 (Waste Water)

Sampled: 04/07/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		3.2	3.0	mg/L	3	B1D0715	04/07/21	04/12/21	SM 5210 B
Specific Conductance (EC)		630	1.0	µS/cm	1	B1D0717	04/07/21	04/07/21	SM 2510 B
Nitrate as N	D2	ND	0.90	mg/L	2	B1D0807	04/08/21	04/08/21	EPA 300.0
Nitrite as N	D2	ND	0.60	mg/L	2	B1D0807	04/08/21	04/08/21	EPA 300.0
pH		7.9	0.10	pH Units	1	B1D0717	04/07/21	04/07/21	SM 4500-H B
Total Kjeldahl Nitrogen		15	0.50	mg/L	1	B1D0903	04/09/21	04/14/21	EPA 351.2
Total Nitrogen		15	0.90	mg/L	2	[CALC]	04/14/21	04/14/21	(CALC)
Total Suspended Solids		ND	4.0	mg/L	1	B1D1305	04/13/21	04/13/21	SM 2540 D

Plant INF. (MD-28S)

HD07011-02 (Waste Water)

Sampled: 04/07/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		160	30	mg/L	30	B1D0715	04/07/21	04/12/21	SM 5210 B
Total Suspended Solids		530	40	mg/L	10	B1D1305	04/13/21	04/13/21	SM 2540 D

SS # Main Well

HD07011-03 (Ground Water)

Sampled: 04/07/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Specific Conductance (EC)		480	1.0	µS/cm	1	B1D1911	04/19/21	04/19/21	SM 2510 B

Notes and Definitions

- BS2 Recovery for this analyte was biased low. Results were accepted based on duplicate results.
- D2 Sample diluted due to high concentration(s) of non-target analyte(s).
- DUP1 A high RPD was observed between a sample and this sample's duplicate.
- MS1 Recovery for this analyte was affected by matrix.
- µg/L micrograms per liter (parts per billion concentration units)
- mg/L milligrams per liter (parts per million concentration units)
- mg/kg milligrams per kilogram (parts per million concentration units)
- ND Analyte NOT DETECTED at or above the reporting limit
- RPD Relative Percent Difference
- Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field.
 If the test was performed in the laboratory, the hold time was exceeded. (for aqueous matrices only)

Madera County Public Works - Special Districts
200 W. 4th Street, 3rd Floor
Madera CA, 93637

Project: MD-28S Ripperdan
Project Number: System Number 2014-0820
Project Manager: Marty Duvall

Reported:
10/18/2021

PLANT EFF.

HI28044-01 (Waste Water)

Sampled: 09/28/21 13:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		14	3.0	mg/L	3	B1I2916	09/29/21	10/04/21	SM 5210 B
Specific Conductance (EC)		710	1.0	µS/cm	1	B1I2920	09/29/21	09/29/21	SM 2510 B
Nitrate as N	D2	ND	0.80	mg/L	2	B1I2912	09/29/21	09/29/21	EPA 300.0
Nitrite as N	D2	ND	0.60	mg/L	2	B1I2912	09/29/21	09/29/21	EPA 300.0
pH		7.9	0.10	pH Units	1	B1I2920	09/29/21	09/29/21	SM 4500-H B
Total Kjeldahl Nitrogen		16	0.50	mg/L	1	B1J0503	10/05/21	10/06/21	EPA 351.2
Total Nitrogen		16	0.80	mg/L	2	[CALC]	10/06/21	10/06/21	(CALC)
Total Suspended Solids		12	5.0	mg/L	1.25	B1J0414	10/04/21	10/04/21	SM 2540 D

PLANT INF.

HI28044-02 (Waste Water)

Sampled: 09/28/21 13:00

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		190	100	mg/L	100	B1I2916	09/29/21	10/04/21	SM 5210 B
Total Suspended Solids		330	40	mg/L	10	B1J0414	10/04/21	10/04/21	SM 2540 D

Main Well

HI28044-03 (Ground Water)

Sampled: 09/28/21 13:15

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Specific Conductance (EC)		480	1.0	µS/cm	1	B1J0520	10/05/21	10/05/21	SM 2510 B
Nitrate as N		5.9	0.80	mg/L	2	B1I2912	09/29/21	09/29/21	EPA 300.0

Notes and Definitions

D2	Sample diluted due to high concentration(s) of non-target analyte(s).
DUP2	RPD for duplicate analysis exceeded limits due to matrix interference.
MS1	Recovery for this analyte was affected by matrix.
MS3	Recovery for this analyte was biased low; associated blank spike recoveries are within range.
RPD	The RPD result exceeded the QC control limits. However, both percent recoveries were acceptable.
RPD2	A high RPD was observed due to the low concentration of the target analyte.
RPD3	The RPD is out of range for this spike and its duplicate due to a low or high bias of one of the two spikes.
µg/L	micrograms per liter (parts per billion concentration units)
mg/L	milligrams per liter (parts per million concentration units)
mg/kg	milligrams per kilogram (parts per million concentration units)
ND	Analyte NOT DETECTED at or above the reporting limit
RPD	Relative Percent Difference
	Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field.
	If the test was performed in the laboratory, the hold time was exceeded. (for aqueous matrices only)

Madera County Public Works - Special Districts
200 W. 4th Street, 3rd Floor
Madera CA, 93637

Project: MD-28S Ripperdan
Project Number: System Number 2014-0820
Project Manager: Marty Duvall

Reported:
11/24/2021

PLANT EFF.

HK09010-01 (Waste Water)

Sampled: 11/09/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		3.9	3.0	mg/L	3	B1K0916	11/10/21	11/15/21	SM 5210 B
Specific Conductance (EC)		700	1.0	µS/cm	1	B1K0915	11/09/21	11/09/21	SM 2510 B
Nitrate as N		1.8	0.80	mg/L	2	B1K1009	11/10/21	11/10/21	EPA 300.0
Nitrite as N		ND	0.60	mg/L	2	B1K1009	11/10/21	11/10/21	EPA 300.0
pH		7.8	0.10	pH Units	1	B1K0915	11/09/21	11/09/21	SM 4500-H B
Total Kjeldahl Nitrogen		6.8	0.50	mg/L	1	B1K1605	11/16/21	11/18/21	EPA 351.2
Total Nitrogen		8.5	0.80	mg/L	2	[CALC]	11/18/21	11/18/21	(CALC)
Total Suspended Solids		ND	4.0	mg/L	1	B1K1019	11/10/21	11/10/21	SM 2540 D

PLANT INF.

HK09010-02 (Waste Water)

Sampled: 11/09/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Biochemical Oxygen Demand		230	100	mg/L	100	B1K0916	11/10/21	11/15/21	SM 5210 B
Total Suspended Solids		250	40	mg/L	10	B1K1019	11/10/21	11/10/21	SM 2540 D

Main Well

HK09010-03 (Ground Water)

Sampled: 11/09/21 09:30

Analyte	Flag	Result	Reporting Limit	Units	Dilution	Batch	Prepared	Analyzed	Method
Inorganics									
Specific Conductance (EC)		510	1.0	µS/cm	1	B1K1518	11/15/21	11/15/21	SM 2510 B
Nitrate as N		6.4	0.80	mg/L	2	B1K1009	11/10/21	11/10/21	EPA 300.0

Notes and Definitions

- BS2 Recovery for this analyte was biased low. Results were accepted based on duplicate results.
- DUP1 A high RPD was observed between a sample and this sample's duplicate.
- MS3 Recovery for this analyte was biased low; associated blank spike recoveries are within range.
- µg/L micrograms per liter (parts per billion concentration units)
- mg/L milligrams per liter (parts per million concentration units)
- mg/kg milligrams per kilogram (parts per million concentration units)
- ND Analyte NOT DETECTED at or above the reporting limit
- RPD Relative Percent Difference
- Analysis of pH, filtration, and residual chlorine is to take place immediately after sampling in the field. If the test was performed in the laboratory, the hold time was exceeded. (for aqueous matrices only)

APPENDIX 3 – WATER BALANCE

	WW Effluent		Infiltration rate				Storage
	Gal/month*	Gal/day**	Ft/day	Gal/day***	storage half full	Gal/day	Increase per day
Jan	41900	1,352	1.02	53,712	0.25	13,428	-12,076
Feb	34700	1,239	1.02	53,712	0.25	13,428	-12,189
March	43500	1,403	1.02	53,712	0.25	13,428	-12,025
April	30900	1,030	1.02	53,712	0.25	13,428	-12,398
May	42700	1,377	1.02	53,712	0.25	13,428	-12,051
June	35400	1,180	1.02	53,712	0.25	13,428	-12,248
July	27200	877	1.02	53,712	0.25	13,428	-12,551
Aug	21300	687	1.02	53,712	0.25	13,428	-12,741
Sept	14400	480	1.02	53,712	0.25	13,428	-12,948
Oct	26800	865	1.02	53,712	0.25	13,428	-12,564
Nov	23300	777	1.02	53,712	0.25	13,428	-12,651
Dec	62300	2,010	1.02	53,712	0.25	13,428	-11,418

Average Daily flow 1,106 gal/day

* Gal/Month = calculated by site operators from pump hours

** Gal/day = Gal/Month /# of days per month

*** Gal/day = Infiltration rate (ft/day) * cross sectional surface area of pits (ft²) *7.48 (ft³/gal)

440 Cross sectional area per leach pit (ft²)

7,040 Cross sectional area of 16 leach pits = 440 * 16 = 7040 (ft²)